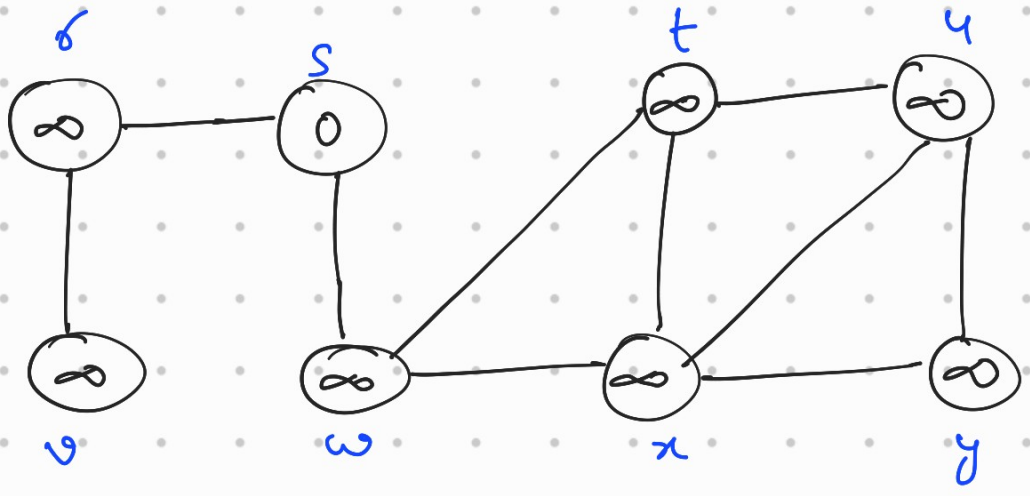
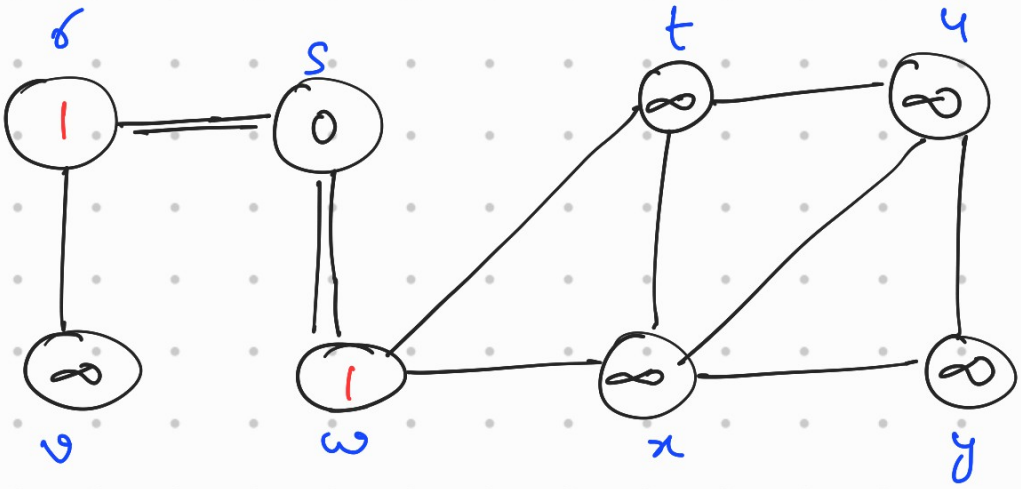
BFS

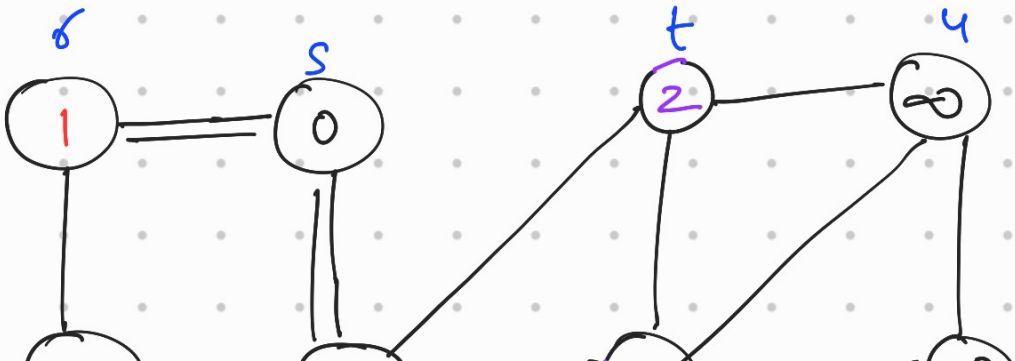


Q | s |
0

(b)

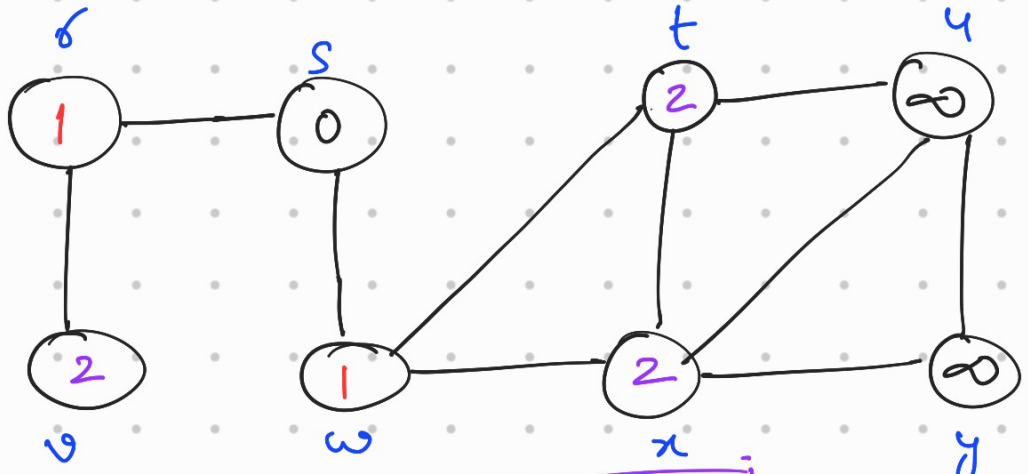


Q | w | r |
1 1

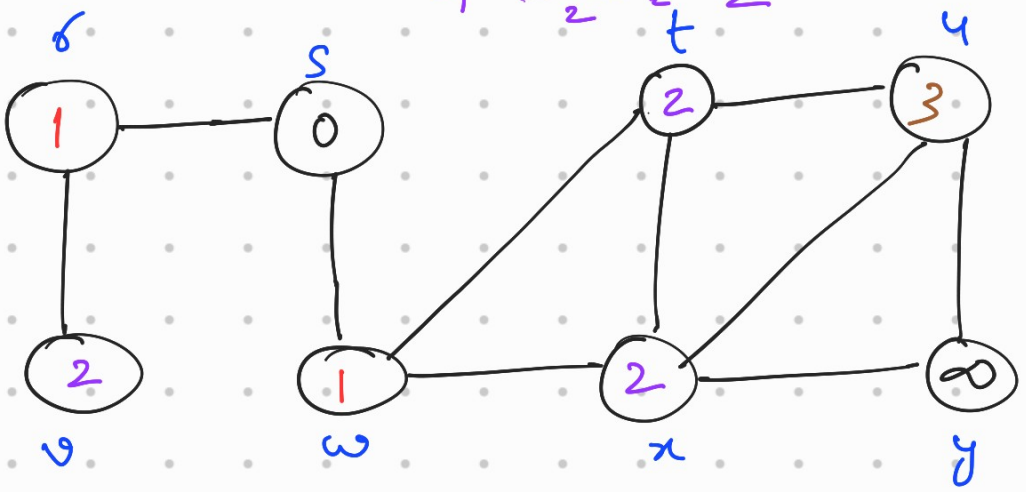




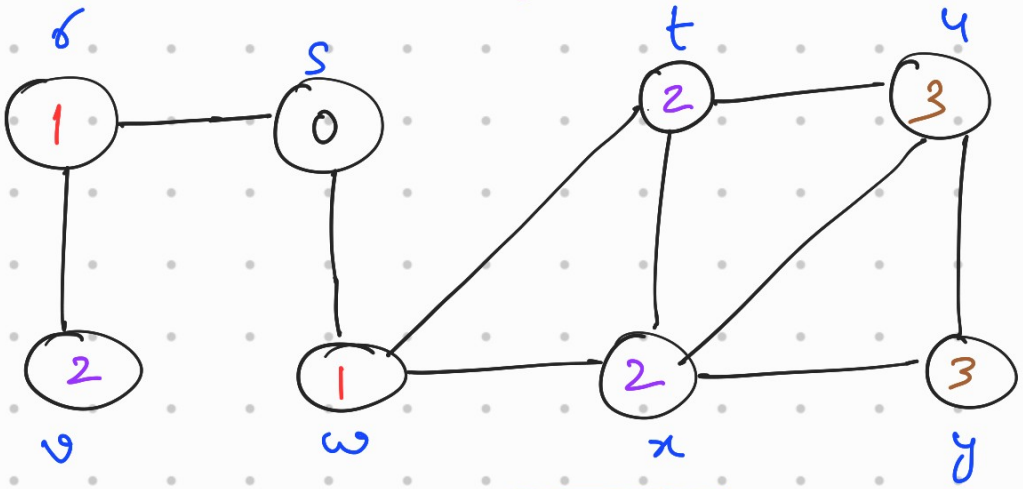
Q | r | t | x |
1 2 2



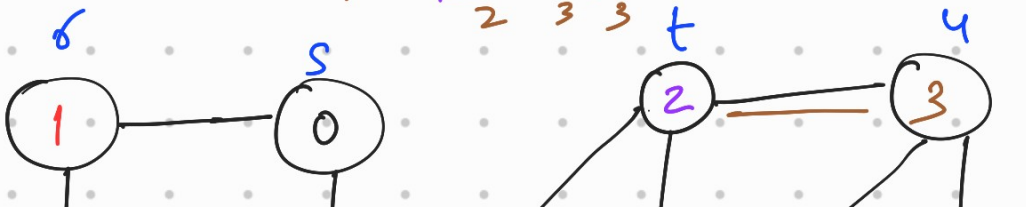
Q | t | x | v |
2 2 2

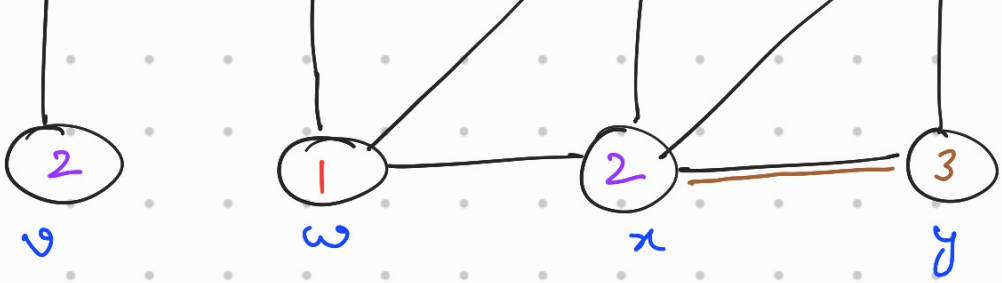


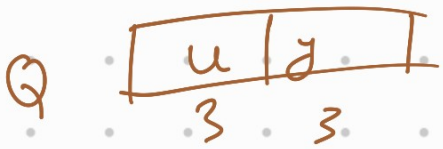
Q | x | v | u |
2 2 3

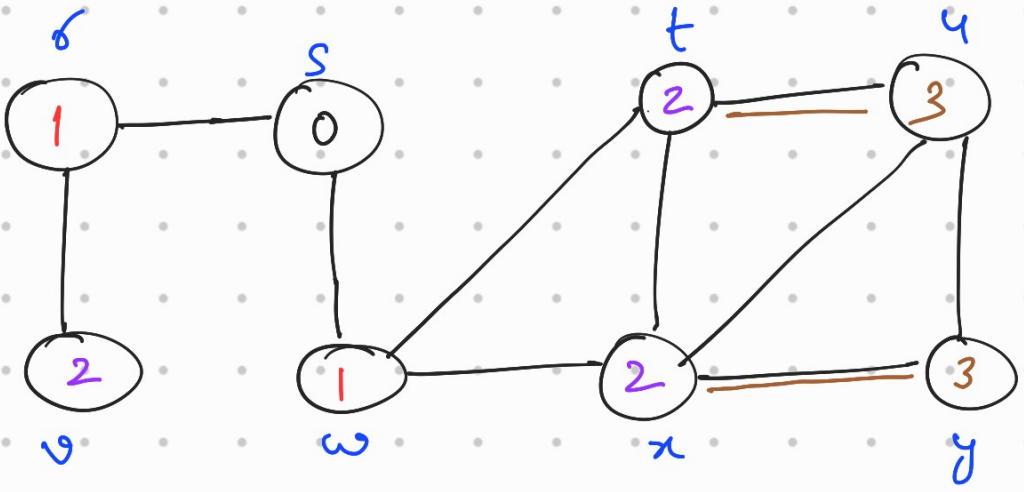


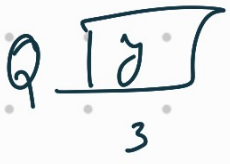
Q | v | u | y |
2 3 3



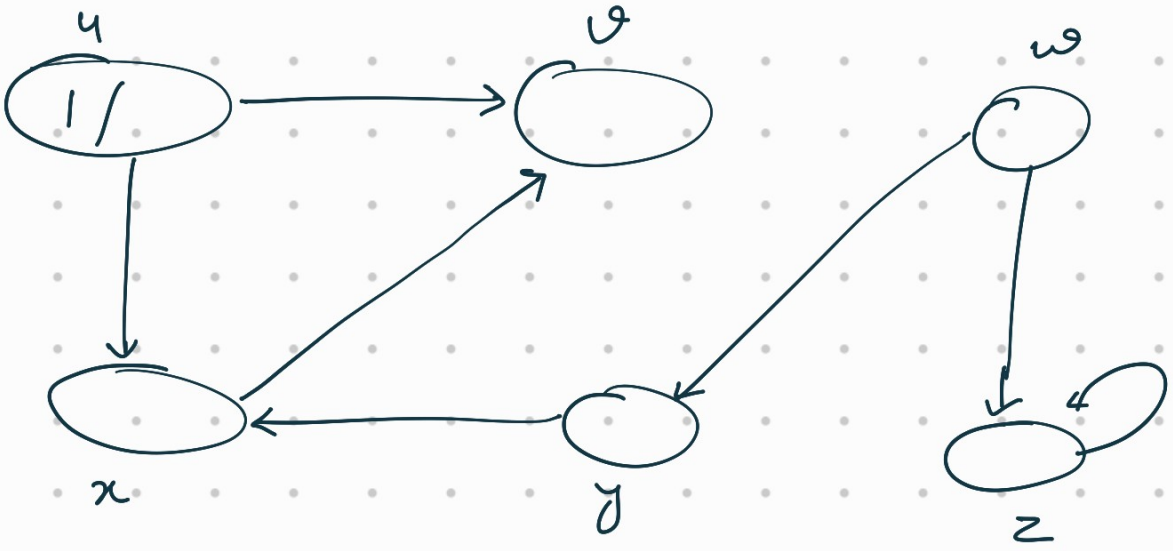








DFS



DFS(G)

```
for each vertex u ∈ G.V
        u.color = WHITE
        u.π     = NIL
time = 0
for each vertex u ∈ G.V
        if u.color == WHITE
                DFS-VISIT(G, u)
```

DFS-VISIT(G, u)

```
time = time + 1
u.d = time
u.color = GRAY
for each v ∈ G.adj[u]
        if v.color == WHITE
                v.π = u
                DFS-VISIT(G, v)
u.color = BLACK
time = time + 1
u.f = time
```

BFS(G, s)

```
for each vertex u ∈ G.V - {s}
        u.color = WHITE
```

```
                u.d   = ∞
                u.π   = NIL
s.color         = GRAY
s.d             = 0
s.π             = NIL
Q               = ϕ
ENQUEUE (Q, s)
  while  Q ≠ ϕ
         u = DEQUEUE (Q)
         for each u ∈ G.Adj[u]
                  if u.color == WHITE
                           v.color = GRAY
                           v.d = u.d + 1
                           v.π = u
                           ENQUEUE (Q, v)
         u.color = BLACK
```